产 品 规 格 书
Product Specification

CUSTOMER	客户:	
CUSTOMER PN	客户 PN:	
HANG CRYSTAL P/N	杭晶物料编码:	**8122S-25.000NQ33FTL**
MODEL	产品型号:	**TCXO SMD 2.5x2.0, Clipped Sine, 3.3V**
NOMINAL FREQUENCY	频率:	**25.000MHz**
ISSUE DATE	日期:	**2022 / 09 / 06**

CUSTOMER'S APPROVAL

客户确认

(PLEASE RETURN A COPY WITH APPOVAL)
（请将确认的复印件返回我司）

APPROVED	QA
MB.	J. Jiang

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Revision	Description / ECN	Prepared	Approved	Date
1	Initial release	MB	James Jiang	2022-09-06
2	Not issued			
3	Not issued			
4	Not issued			

1. NOMINAL AND MAXIMUM RATINGS, OPERATING AND STORAGE CONDITIONS

	PARAMETER	SYMB.	MIN	TYP	MAX	Unit	Conditions / Remarks
1	Nominal frequency	F_N		**25.000**		MHz	--
2	Maximum supply voltage	V_{MAX}	–0.3		+4.6	V_{DC}	Between V_{CC} and GND
3	Nominal supply voltage	V_{CC}	+3.135	+3.30	+3.465	V_{DC}	--
4	Output load (resistive)	RL	9	10	11	kΩ	Clipped sine wave output
5	Output load (capacitive)	CL	9	10	11	pF	Clipped sine wave output
6	Operating temperature range	T_{OP}	–30	+25	+85	°C	Note 1
7	Storage Temperature Range	T_{ST}	–40		90	°C	--
8	Enable / Disable function	E/D	Not available				Pin 1 GND

Note 1: Unit stays within all relevant parameter limits as specified under point 2

2. ELECTRICAL PARAMETER LIMITS

	PARAMETER	SYMB.	MIN	TYP	MAX	Unit	Conditions / Remarks
1	Frequency calibration	$\Delta f/F_N$	–1.0		+1.0	ppm	Offset from nominal at +25°C
2	Frequency stability over T_{OP}	$\Delta f/F_{OP}$	–0.5		+0.5	ppm	Over T_{OP}; Note 1
3	Frequency VS voltage changes	$\Delta f/F_V$	–0.2		+0.2	ppm	Vcc ±5% at +25°C
4	Frequency VS load changes	$\Delta f/F_L$	–0.2		+0.2	ppm	RL//CL ±5% at +25°C
5	Frequency VS reflow	$\Delta f/F_{RFL}$	–1.0		+1.0	ppm	Max. two times reflow after 24h
6	Hysteresis	Δf	–0.6		+0.6	ppm	---
7	Aging first year	$\Delta f/F_{A1}$	–1.0		+1.0	ppm	at +25°C
8	Aging over 2 years	$\Delta f/F_{A2}$	–1.5		+1.5	ppm	at +25°C
9	Output amplitude voltage level	V_{P-P}	0.8	1.1		V_{AC}	Clipped sine wave DC coupled
10	Phase noise at 10Hz offset	L_{RMS}			–80	dBc/Hz	at +25°C
	at 100Hz offset	L_{RMS}			–112	dBc/Hz	at +25°C
	at 1kHz offset	L_{RMS}			–133	dBc/Hz	at +25°C
	at 10kHz offset	L_{RMS}			–143	dBc/Hz	at +25°C
	at 100kHz offset	L_{RMS}			–150	dBc/Hz	at +25°C
11	Current consumption	I_{CC}			2.0	mA	Under load RL//CL ±10%
12	Startup time	t_{STRT}			2.0	ms	V_{P-P} reach >90% of amplitude

Note 1: Referenced to midpoint between minimum and maximum frequency over specified temperature range.

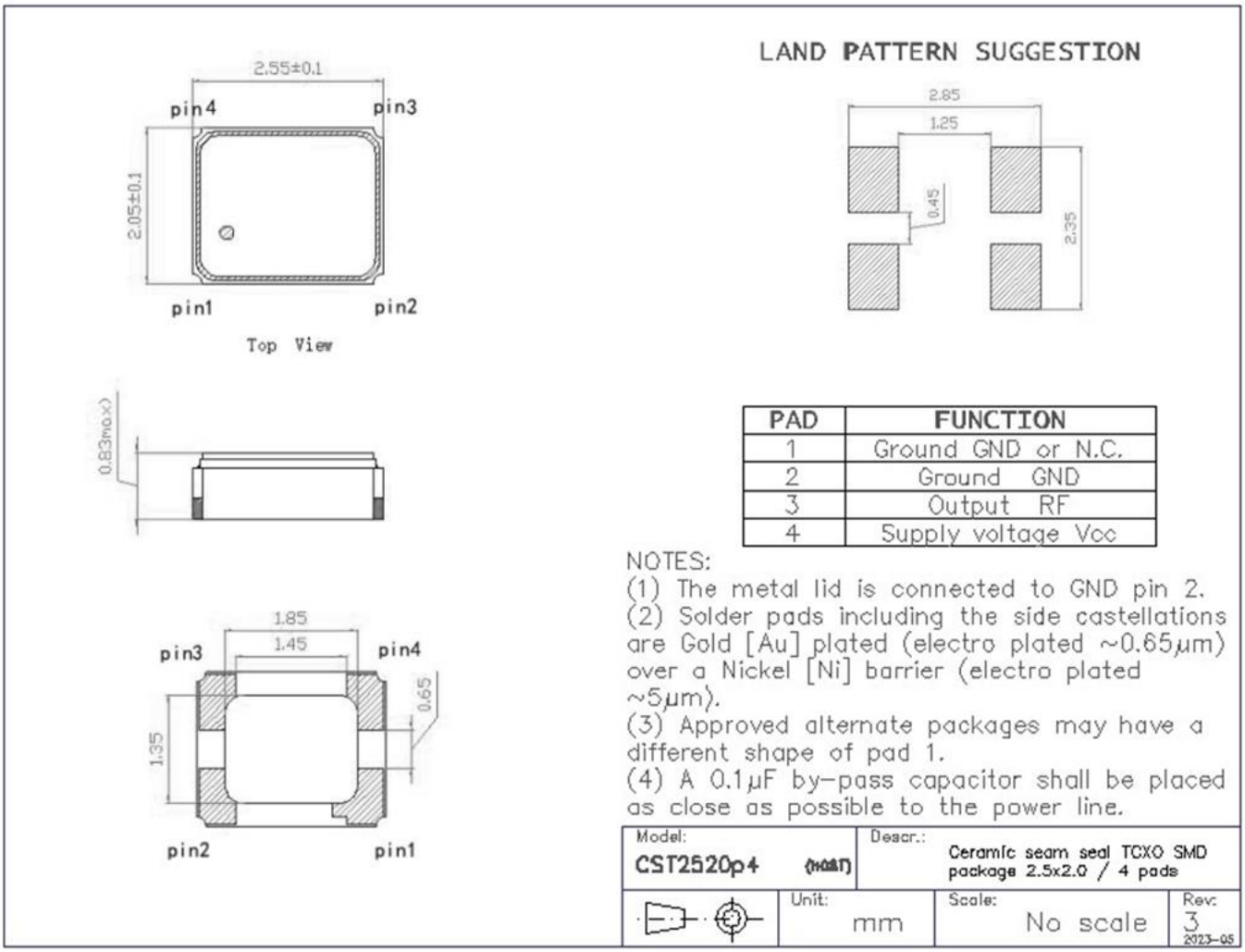
3. PRODUCT MARKING

1	**FF.fff**	Nominal frequency in MHz (three digits after decimal point)											
2	**HCI**	Company logo											
3	**Y**	Year code of manufacturing (see table below)											
	Year	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032
	Code	V	W	X	Y	Z	A	B	C	D	E	F	G
4	**M**	Month code of manufacturing (see table below)											
	Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
	Code	A	B	C	D	E	F	G	H	J	K	L	M

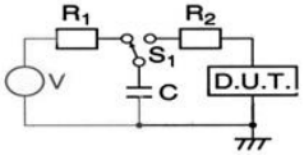
HCI YM
• FF.fff

4. OUTLINE DRAWING

	Package description	Package model	Remarks
1	Ceramic seam seal SMD package 2.5x2.0mm with 6 pads for TCXO	CST2520p4sph80	--



5. RELIABILITY TEST INFORMATION

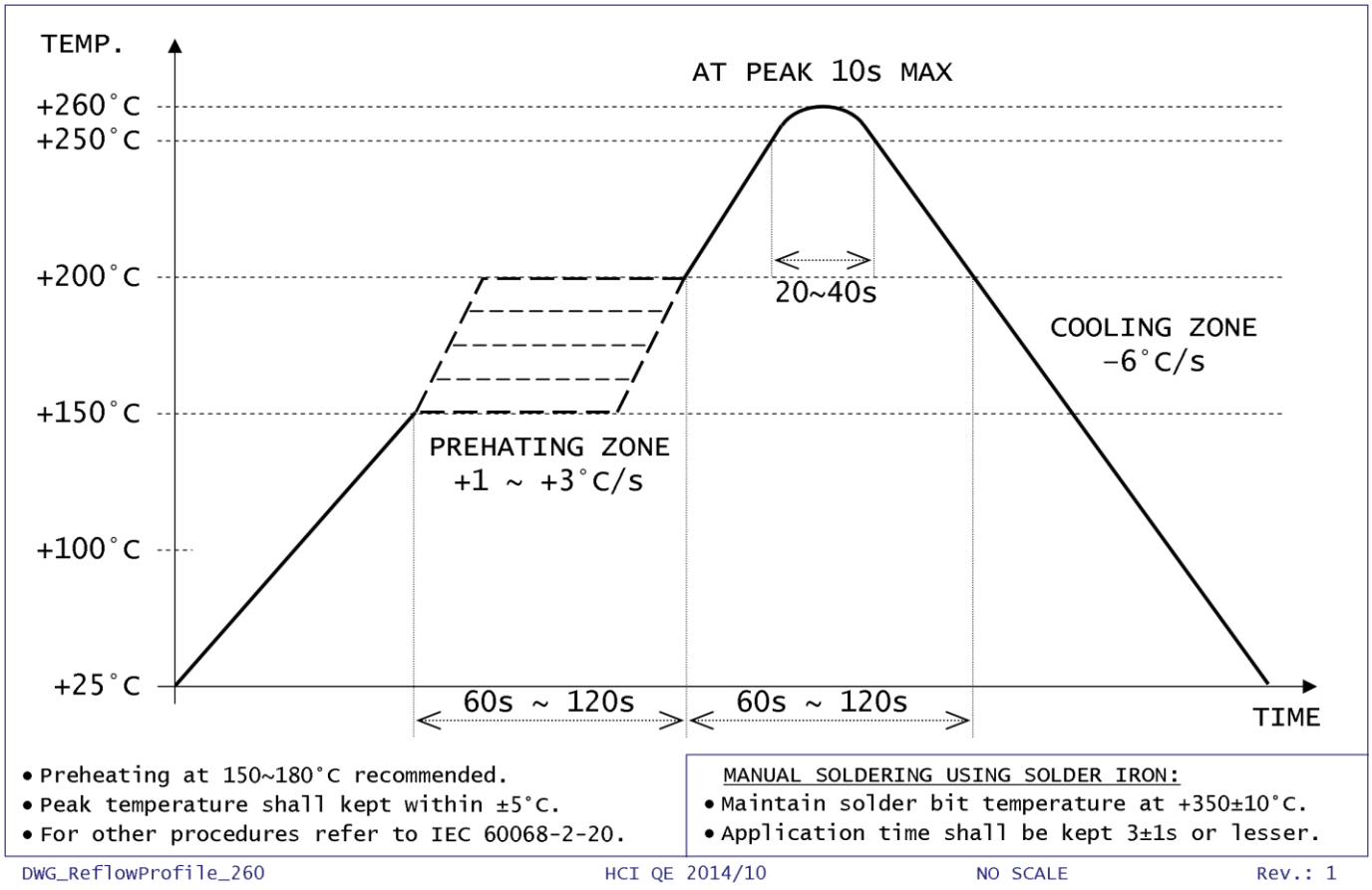
	Test item	Test conditions	Criteria
1	High temperature storage	Temperature: +85°C ±5°C Time: 240 ±4 Hours Tested after 4 to 12h at room temperature.	±1.0ppm
2	Low temperature storage	Temperature: –40°C ±5°C Time: 240 ±4 Hours Tested after 4 to 12h at room temperature.	±1.0ppm
3	Temperature humidity bias THB	Temperature: +85°C ±5°C Humidity: 85% ±5% RH Time: 240 ±4 Hours BIAS: Supply Voltage Tested after 4 to 12h at room temperature.	±1.0ppm
4	Temperature cycling	Low Temp. cycle: –40°C ±2°C High Temp. cycle: +85°C ±5°C Time: 30min each cycle Number of cycles: 1,000 Tested after 4 to 12h at room temperature.	±1.0ppm
5	Aging	Temperature: +85°C ±5°C Time: 30d Tested after 4 to 12h at room temperature.	±1.0ppm
6	Resistance to solder heat	Reflow peak temp.: +260°C ±5°C (refer to rec. profile) Number of cycles: 3 times Tested after 4 to 12h at room temperature.	±1.0ppm
7	Solderability (MIL-STD-883E)	Dip in flux: 5~10 Seconds Temperature: 230℃±10℃ Time: 5 Seconds Tested after 4 to 12h at room temperature.	>95% cover.
8	Drop test	Drop height: 120cm Number of cycles: 12 times Drop height: 150cm Number of cycles: 9 times With jig (120~150g) onto iron plate Tested after 24h at room temperature.	±1.0ppm
9	Vibration	Frequency Range: 20~2000Hz PSD: 0.053g^2 Time: 40min each direction (X,Y,Z) Tested after 4 to 12h at room temperature.	±1.0ppm
10	ESD-HBM	HBM, V=±1KV, C=100pF, R1=10M, R2=1.5K, 3times 	±1.0ppm

6. ENVIRONMENTAL COMPLIANCE INFORMATION

		Compliance information
1	RoHS	This product is fully RoHS compliant, 6/6 compliant per EU legislation.
2	RoHS 2	In regards of RoHS 2, CE marking directive for finished products, we can provide RoHS test reports and MDS to show compliance, but since our product is not a final application we have no CE mark.
3	Lead-Free	This product is considered Lead-Free, Lead (Pb) contamination is controlled to be below 200ppm.
4	Halogen-Free	This product is compliant to IEC 61249-2-21:2003 (Br<800ppm / Cl<800ppm).
5	REACH (SVHC)	This product does not contain substances (SVHC) listed by REACH, we continuously monitor updates of the list of SVHC's
6	PFOS / PFOA Free	This product is free of any PFOS / PFOA.
7	Electrostatic Discharge (ESD) sensitivity	This product is ESD sensitive and requires precautions for handling and storage. Follow JEITA EIAJ ED-4701 or JSD22 or ANSI-ESD-S20-20 or IEC 61000-4-2.
8	Moisture Sensitivity	This product is hermetically sealed and does NOT fall under the classification of moisture sensitivity per J-STD-020C (Standard is for non-hermetically sealed components). If required we suggest to use LEVEL 1

7. RECOMMENDED SOLDERING INFORMATION

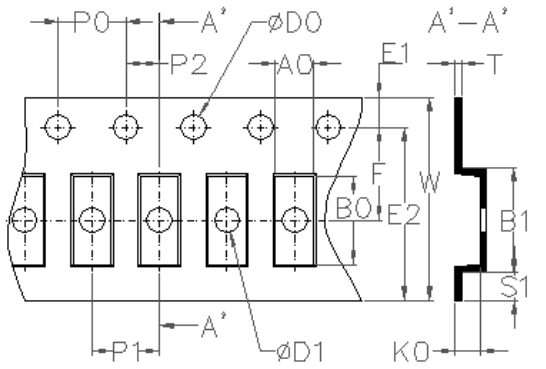
RECOMMENDED REFLOW SOLDER PROFILE – PEAK TEMPERATURE UP TO +260°C



8. PACKAGING

Carrier

	Parameter	STANDARD PACKAGING	ALTERNATE PACKAGING
1	A0	2.3±0.1	
2	B0	2.8±0.1	
3	K0	1.15±0.1	
4	B1	3.4±0.1	
5	P0	4.0±0.1	
6	P1	4.0±0.1	
7	T	0.3±0.05	
8	W	8.0±0.2	

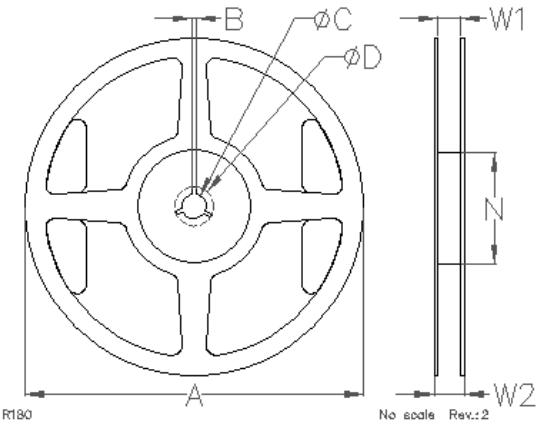


Note 1: All dimensions in [mm].
Note 2: All dimensions not specified or not being shown follow EIA-481 standard.

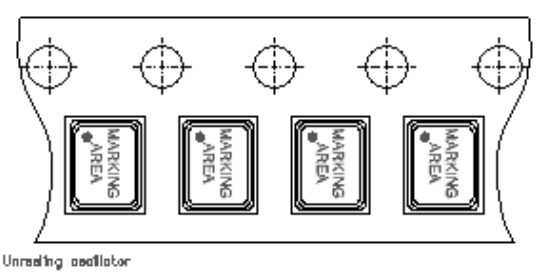
Reel

QTY per reel: 3,000pcs MAX

	Parameter	STANDARD PACKAGING	ALTERNATE PACKAGING
9	A	$180^{+0}_{-1.5}$	
10	B	2.0±0.5	
11	ØC	13±0.2	
12	ØD	21±0.8	
13	N	60±2	
14	W1	9.0±0.3	
15	W2	11.4±1.0	



Note 1: All dimensions in [mm]. Dimension W1 is measured near the Hub (N).
Note 2: All dimensions not specified or not being shown follow EIA-481 standard.

Unreeling information

	Oscillator product's orientation
16	This product is a polarized component which requires a certain orientation; Pin 1 is identified on top side marking with a DOT. In the carrier tape is the component oriented with pin 1 towards the sprocket holes. (per EIA-481)